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Repository: <https://github.com/nklloyd/flyprecise>

# Alpha Build

Fly Precise

## Usability

### Interface

Data displays consisting of volumetric output from the flow meter on an LCD display. The flow meter is connected to an arduino program that outputs the information on the LCD display. An additional flow meter is connected with a built-in display that displays the volumetric output data for redundancy purposes in the testing phase. The proximity sensor is similarly connected to an arduino display with an LCD display output to determine the specific gravity of the liquid via the height of the hydrometer.

### Navigation

The navigable components consist of informational sensors and control pumps and valves. The proximity sensor determines the specific gravity and density of the liquid, while the flow meter calculates the volumetric output of the pump. These systems are easily navigable, connecting to the arduino program and outputting the result on an LCD display. This work is done in the background and the user will not have to navigate it much, with the curated data (volume and density) output directly on the screen. The control valves have a motor apparatus which will be controlled by the servo motors and the arduino program. For testing purposes, the valves also have manual handles with "OFF" and "ON" positions. The pump is currently controlled by the Power Supply equipment to supply a controlled voltage. This is the most complex for navigability in the testing phase, but the control of this will ultimately be completed by the arduino program in the background while the user simply inputs a mass value to the interface.

### Perception

The alpha build requires more interaction than the final product is desired to have. The main user interaction includes two valves where the user has to control liquid flow that will later be electrically controlled by motors. The purpose of the artifact is to reliably measure volume by flow and density by specific gravity. Currently our results are reliably displayed on a serial plot using a laptop but our plans are to implement an LCD screen for simpler data readings and a possible upgrade to the screen later on.

## Responsiveness

The will save resources by running display and sensor code only during the pumping state. The sensor will acquire the density of the fuel while it's in the bypass and send the data to the main arduino which will display density and total volume of fuel inside of the secondary test bucket. The display will be updated every second and save resources at the end by detecting the pump state. In the future if api calls are made to a website or application then a buffering system will be implemented to only send information that is different from the previously entered data.

## Build Quality

### Robustness

The physical robustness of the system is sound, as the custom 3D-printed components were designed to interface appropriately with the off-the-shelf PVC parts and fixtures. There are no leaks in the fluid system. As for the software control of the system, the processing of the flow-meter data and the distance sensor data within the hydrometer assembly is smooth. From here, the display of this data sees no glitches or malfunctions.

### Consistency

The design produces consistent output for mass. Regardless of external factors such as air pressure or temperature, the fuel mass output from the device will match the user's input. This improves upon existing systems where the mass output differs from the intended amount due to variations in external factors.

### Aesthetic Rigor

The fuel line is currently functional and any changes from the current piping will be solely for material changes. The photos of previous designs will show no major aesthetic changes to the modern fuel line mockup. The future iterations of the design will feature visually smaller components (hydrometer, density chamber, minor changes from L to T junctions) and shorter lines for compact integration to a fuel line. Additions will come from calorimeters which may add significant length or a secondary bypass to the current fuel line and change the aesthetics of the project.

# Vertical Features

## External Interface

Both main use cases already have a working interface. Density readings show on the LCD, and flow rate and total volume show on the Serial Monitor. Neither one is connected to a data store yet, so the readings are only live and nothing gets saved. Later, both sides will feed into the same database, and the app will show density, volume, and pounds together for each fueling run.

## Persistent State

The prototype doesn't store data yet. The flow total resets to 0 when the board loses power, and density readings only show live on the LCD, so for alpha testing we're recording readings by hand into our results sheet. Next, we plan to save the density and flow calibration values to a database, each run's density, volume, and pounds will be sent over wifi to a database, and a companion app will display the live readings and past fueling records.

## Internal Systems

Two systems presently implemented for data processing. One system connects to the flow meter wiring to determine volumetric output in an arduino program and connect it to an LCD display. The second system connects to a proximity sensor to determine the height of the hydrometer above the fluid